**Unofficial worked solution** — written by UWMathFinder, not by the instructor. Spot an error? Report it on the site.

### Math 208 Midterm 2, Winter 2024 (Yang Zhang) — Problem 1

From $B$ the pivots are in columns 1 and 2.

(1) Row space basis: the nonzero rows of $B$, $\{(1,-1,1,1),\ (0,1,2,1)\}$; $\boxed{\dim \operatorname{row}(A) = 2}$.

(2) Column space basis: the pivot columns of $A$, $\left\{ \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ -2 \\ 0 \end{bmatrix} \right\}$; $\boxed{\dim \operatorname{col}(A) = 2}$.

(3) $\boxed{\operatorname{rank}(A) = 2}$.

(4) $Bx = 0$: $x_2 = -2x_3 - x_4$, $x_1 = x_2 - x_3 - x_4 = -3x_3 - 2x_4$. With $x_3 = s$, $x_4 = t$:

$$\operatorname{null}(A) = \operatorname{span}\left\{ \begin{bmatrix} -3 \\ -2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -2 \\ -1 \\ 0 \\ 1 \end{bmatrix} \right\}, \qquad \boxed{\operatorname{nullity}(A) = 2}.$$

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**Math 208 Midterm 2, Winter 2024 (Yang Zhang) — Problem 2**

(1) **No**: four vectors in $\mathbb{R}^3$ are always linearly dependent, so they can't be a basis.

(2) Row reduce $[A \mid I]$ with $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 2 & 3 \end{bmatrix}$: subtract row 1 from rows 2 and 3, then row 2 from row 3, then clear upward:

$$A^{-1} = \boxed{\begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{bmatrix}} \quad (\text{check } AA^{-1} = I).$$

(3) $\mathbf{x} = A^{-1}\mathbf{b} = A^{-1}\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \boxed{\begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix}}$ (check: $2\mathbf{u}_1 - \mathbf{u}_2 = \mathbf{b}$).

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### Math 208 Midterm 2, Winter 2024 (Yang Zhang) — Problem 3

(1) A basis for the plane $x - y + 2z = 0$ is $(1, 1, 0)$, $(-2, 0, 1)$. Let

$$T(\mathbf{x}) = \begin{bmatrix} 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \mathbf{x}.$$

Its range is the span of its columns, which is exactly the plane. It is not one-to-one because $T(\mathbf{e}_3) = \mathbf{0}$ (any map $\mathbb{R}^4 \to \mathbb{R}^3$ has a nonzero kernel).

(2) Multiply on the left by $D + XB$: $XA = (D + XB)C = DC + XBC$. So $X(A - BC) = DC$ and

$$\boxed{X = DC(A - BC)^{-1}}.$$

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**Math 208 Midterm 2, Winter 2024 (Yang Zhang) — Problem 4**

(1) $T(\mathbf{x}) = \mathbf{x} \iff (A - I)\mathbf{x} = \mathbf{0}$ with $A - I = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 1 \\ 1 & 0 & 1 \end{bmatrix}$, giving $x_3 = 0$, $x_1 = 0$, $x_2$ free:

$$S = \operatorname{span}\{(0, 1, 0)\}, \qquad \boxed{\dim S = 1}.$$

(2) Expand along row 3 (only entry: $a_{34} = 1$, sign $(-1)^{3+4} = -1$):

$$\det A = -\det \begin{bmatrix} -1 & 1 & 1 \\ -1 & -1 & 0 \\ 1 & 0 & 0 \end{bmatrix} = -\Big(1 \cdot \det \begin{bmatrix} 1 & 1 \\ -1 & 0 \end{bmatrix}\Big) = -1.$$

$\boxed{\det A = -1, \quad \det(A^{-1}) = -1}$.

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### Math 208 Midterm 2, Winter 2024 (Yang Zhang) — Problem 5

The four input vectors form a basis of $\mathbb{R}^4$ (their determinant is $4 \neq 0$). Their images $(0,-1,0,0)$, $(0,0,0,1)$, $(0,0,-1,0)$, $(1,0,0,0)$ are also linearly independent. A vector in the kernel is a combination $c_1\mathbf{v}_1 + \cdots + c_4\mathbf{v}_4$ with $c_1T(\mathbf{v}_1) + \cdots + c_4T(\mathbf{v}_4) = \mathbf{0}$, which forces all $c_i = 0$. So the kernel is $\{\mathbf{0}\}$ and **yes, $T$ is one-to-one**.